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Research Article

High Volumetric Energy Density Supercapacitor of Additive-Free Quantum Dot Hierarchical Nanopore Structure

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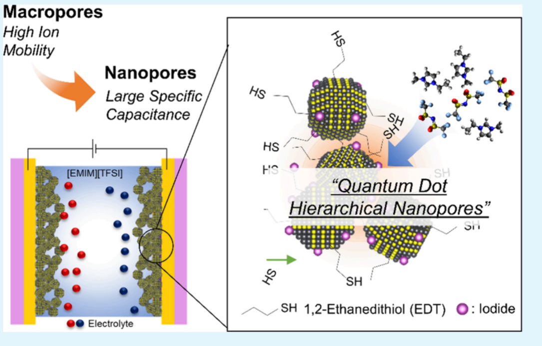
ABSTRACT: The high surface-area-to-volume ratio of colloidal quantum dots (QDs) positions them as promising materials for high-performance supercapacitor electrodes. However, the challenge lies in achieving a highly accessible surface area, while maintaining good electrical conductivity. An efficient supercapacitor demands a dense yet highly porous structure that facilitates efficient ion−surface interactions and supports fast charge mobility. Here we demonstrate the successful development of additive-free ultrahigh energy density electric double-layer capacitors based on quantum dot hierarchical nanopore (QDHN) structures. Lead sulfide QDs are assembled into QDHN structures that strike a balance between electrical conductivity and efficient ion diffusion by employing meticulous control over inter-QD distances without any additives. Using ionic liquid as the electrolyte, the high-voltage ultrathin-film microsupercapacitors achieve a remarkable combination of volumetric energy density (95.6 mWh cm^{-3}) and power density (13.5 W cm^{-3}). This achievement is attributed to the intrinsic capability of QDHN structures to accumulate charge carriers efficiently. These findings introduce innovative concepts for leveraging colloidal nanomaterials in the advancement of high-performance energy storage devices.



KEYWORDS: *supercapacitors, colloidal quantum dots, hierarchical nanopores, volumetric energy density, electric double layer*

1. INTRODUCTION

The surge in mobile electronic systems, such as electric vehicles, memory backup systems, and wearable electronics, has spurred the advancement of energy storage materials, including supercapacitors.[1−3] Consequently, there is a pressing need to enhance the energy density of supercapacitors to meet the demands of these evolving technologies. In the case of electric double-layer capacitors (EDLCs), their capacitance arises solely from the electrostatic interaction between charges and ions at the electrode and electrolyte interface.[4] Hence, the key to achieving a high-performance supercapacitor lies in designing an electrode that provides an extensively accessible surface for efficient ion interaction, enabling rapid charging and discharging capabilities.

The future advancement of supercapacitors will rely on three pivotal factors: nanostructuring the electrode materials to quantum-level size, optimizing their functionalities, and refining the electrode−electrolyte combination to introduce additional capacitance contributions.[5−8] To date, the enhancement of supercapacitor electrodes using nanostructured materials has primarily focused on expanding the surface area or enhancing the accessibility of the material, directly influencing the capacitance.[9] However, these approaches encounter limitations regarding nanostructure size reduction and the electrolyte's ability to effectively penetrate and interact with the innermost regions of the nanostructured surface.[10] Consequently, EDLCs are still considerably short in terms of energy density and volumetric capacitance (C_V) compared to the achievements seen in pseudocapacitors (i.e., C_V > 300 F cm^{-3}).[11−14] Considerable efforts have been dedicated to addressing this challenge in electrode materials for EDLCs, including porous carbon, MXene, 2D materials, and various carbon allotropes like nanotubes and graphene, which possess large specific surface areas.[15−18] Notably, a hierarchical nanoporous structure can be achieved to a certain extent in 2D materials by implementing a strong acidic etching process.[19] However, the inherently 2D nature of these materials restricts deep ion penetration and absorption, posing limitations. To overcome these limitations, precise control over the assembly of nanomaterials to create nanoporous structures is paramount. Exploring innovative strategies to obtain nanoporous structures effectively represents a critical avenue

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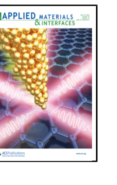

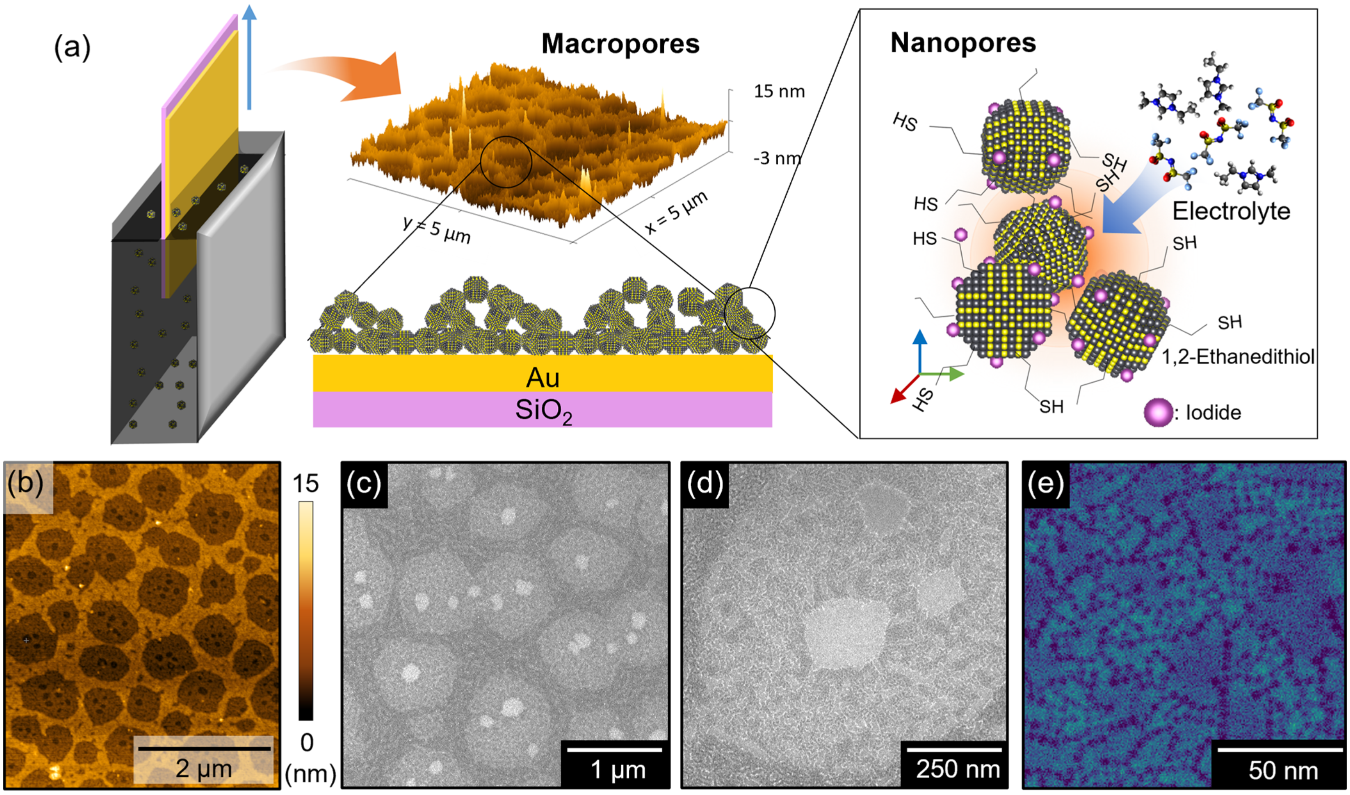

Figure 1. (a) Schematic illustration of the macro-to-nano hierarchical porous structure of QD using dip coating. (b) AFM image and (c−e) FESEM images of an IL pristine sample showing the microporous changes to the nanoporous structure. False color is applied for readability.

for advancing supercapacitor technology to unprecedented levels.

As nanomaterials, colloidal quantum dots (QDs) possess flexibility and low-cost advantages in solution-based processing. Additionally, they exhibit unique size-dependent properties stemming from their 0D nature. QDs, being low-dimensional, have a vast surface-to-volume ratio, allowing for substantial ion accumulation on their surfaces. This accessible surface, in principle, holds the potential for enhancing the behavior of EDLCs once a high loading of QDs on an electrode is achieved. However, most nanoparticle- and QD-based electrodes still face low loading and packing density challenges, adversely affecting their conductivity. Consequently, some additive materials, i.e., conductive agents or binder materials, are typically needed.[20] Unfortunately, including these additive materials compromises the advantageously high surface-to-volume ratio of QDs. Conversely, a densely packed assembly of QDs would impede ion transport, which is unfavorable for supercapacitor operation. Despite their potential, this dilemma has hindered the development of QDs as materials for supercapacitors. In order to enable QDs for high-performance EDLC supercapacitors, it is crucial to achieve a balance between high mass loading and porous structures.

Here we report advanced EDLCs, leveraging a unique architecture of colloidal semiconducting QDs assembled into hierarchical nanopores for electrode construction. This configuration achieves an extraordinary volumetric energy density of 95.8 mWh cm^{-3} (1.43 Wh cm^{-3}), exceeding many contemporary supercapacitor materials. This innovative approach utilizes layer-by-layer (LbL) dip coating[21] and ligand exchange, creating porous QD structures, thereby negating the need for an additional conductive substance. The quantum dot hierarchical nanopore (QDHN) design optimally exposes the QDs to electrolyte ions, harnessing the high surface-to-volume ratio for remarkable energy density in the EDLC. Notably, despite changes in the QDHN electrode thickness, the volumetric energy density remains consistent, demonstrating the future scalability potential of this QD-based supercapacitor and encouraging further exploration of this class of materials for advanced energy storage devices.

2. MATERIALS AND METHODS

2.1. Materials. After synthesis protocol customization, high-quality lead sulfide (PbS) QDs with 4.1 nm diameter were tested and procured from Quantum Solutions (www.quantum-solutions.com). The anhydrous solvents used were acetonitrile (ACN) and methanol (MeOH). 1,2-Ethanedithiol (EDT) and ammonium iodide (NH_4I, 99.999%) were procured from Sigma-Aldrich. The QDs, solvents, and molecules were always stored inside N_2-filled gloveboxes.

2.2. Electrode Fabrications. The PbS QDs were deposited using a LbL dip-coating method on a 30 nm gold-coated SiO_2 substrate adapted from our previous work.[21] The dip-coater (SDI Micro) can perform ultraslow dipping of 1 nm s^{-1}. The QDHN was initially assembled from oleic acid-capped PbS QDs (d = 4.1 nm) in toluene with a concentration of 7 mg mL^{-1}. Subsequent dipping exchanged the oleic acid ligand with EDT (1%, v/v), followed by another dipping for iodide treatment in NH_4I (10 mM) and then MeOH washing. The pulling speed of the EDT dipping was 50 nm s^{-1}. The iodide treatment was done by dipping the EDT-capped PbS film in the NH_4I bath for 30 s before the subsequent pulling. After the MeOH washing (10 s), the sample is dried by ambient air for 30 s before deposition of the next layer. After deposition of all layers, the sample was baked at 80 °C for 1 h. All deposition processes were performed inside a N_2-filled glovebox (Tr. O_2 < 1 ppm; Tr. H_2O < 1 ppm).

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2.3. Material Characterizations. The morphology of the QDHN samples was characterized by using transmission electron microscopy (TEM; JEOL 1230), field-emission scanning electron microscopy (FESEM; Hitachi Hightech, Regulus), and atomic force microscopy (AFM; Hitachi AFM5100N, DFM mode). Optical characterization of the samples was performed using a Fourier transform Infrared (FTIR) spectrophotometer (Shimadzu IRSpirit-T, equipped with QATR-S) and a UV−vis−NIR absorption spectrophotometer (Shimadzu UV-3600 Plus).

2.4. Electrochemical Measurement. Electrochemical measurement was performed using an electrochemical workstation in three- and two-electrode measurements (AMETEK, VersaSTAT4000). The three-electrode measurement used a platinum wire as the counter electrode and Ag/Ag^+ as the reference electrode. All measurements were performed inside an N_2-filled glovebox (Tr. O_2 < 1 ppm; Tr. H_2O < 1 ppm). Before use, the 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][TFSI]) ionic liquid (IL) was utterly dried to avoid any protic effect. The capacitance values were calculated from cyclic voltammetry (CV) measurement with a scan rate of 10−500 mV s^{-1} for the three-electrode measurement. The potential window (ΔV) of 0 V to −1 V vs Ag/Ag^+ was used in the three-electrode measurement to ensure that no redox reaction occurred, either on the current collector or on the QDHN itself. The specific capacitance of the QDHN in the three-electrode measurement was obtained by integrating the current response (I).

$$C = \frac{1}{\nu R \Delta V} \int I \, dV \tag{1}$$

Here, ν represents the voltage scan rate, while R represents the area (A), volume, or mass of the film. The volume of the film was approximated by the thickness measured by AFM measurement. The QDHN mass was determined by multiplying the assumed volume of the film by the mass density of PbS (ρ = 7.3 g cm^{-3}). In addition, the specific capacity was also calculated from the galvanostatic charge−discharge (GCD) curve using the following relationship:

$$Q = I\Delta t \tag{2}$$

Here, I and Δt are the discharging current and discharging time, respectively. Electrochemical impedance spectroscopy (EIS) measurements were performed under a frequency range of 100 mHz to 100 kHz. The volumetric energy density (E_{vol}, mWh cm^{-3}) and volumetric power density (P_{vol}, W cm^{-3}) of the QDHN microsupercapacitor were determined from the two-electrode measurement and calculated by using the following equations:[22]

$$E_{vol} = \frac{\int_0^V IV(t)\, dt}{At} \tag{3}$$

$$P_{vol} = \frac{E_{vol}}{\Delta t} \times 3.6 \tag{4}$$

3. RESULT AND DISCUSSION

3.1. Hierarchical Nanoporous Assemblies of Colloidal QDs. To maximize the inherent potential of QDs, we manipulated their vast surface area by creating a hierarchical structure containing nanopores. We utilized PbS colloidal QDs as our primary building blocks in fabricating QDHNs. PbS was selected because its uniform size and adjustable ligand enable the desired structure to be built. The QDHN was deposited on a patterned gold thin-film electrode using the established LbL dip-coating method (Figure 1a),[21] with further refinement of the deposition process. The insulating oleic acid ligand native to the QDs was replaced by the bidentate EDT, a short ligand molecule adept at connecting the QDs.[21] This ligand exchange, conducted in tandem with the dip-coating process, rendered an additive-free deposition method because charge conduction was managed purely by QD coupling control. The EDT molecules also covalently connect the first layer of QDs to the gold substrate, acting as a support binder. This method is distinct from many supercapacitor electrodes that demand separate conductive matrixes and polymeric binders.[20,23] Each deposition of the PbS QDs layer underwent the ligand exchange process, followed by treatment with NH_4I to passivate any remaining PbS QDs surface bonds and to mitigate charge trap formation.[24−27] Each layer was then cleansed of residuals before deposition of the following layer. The final supercapacitor device was assembled by sandwiching the liquid electrolyte between two PbS QDHN electrodes.

The developed PbS QDHN design hinges on two vital characteristics. The first is the hierarchical porous structure with a maximized surface-to-volume ratio, achieved by arranging 4-nm-diameter QDs, with the interdot distance mainly governed by the EDT connecting ligand length. With smaller QD diameters comes a larger surface-to-volume ratio. Hence, 4-nm-diameter PbS QDs are chosen for their spherical-like form, distinguishing them from other cuboctahedral entities.[28] The second key is immediate iodide treatment on each layer of the pore assemblies. This process effectively passivates the remaining dangling bonds on the PbS QDs that are not connected to any EDT ligands. Notably, this iodide treatment has no detrimental impact on the bridges between the PbS QDs and EDT ligands, preserving the integral structure of the QDHN (Figure S2).

Parts b−e of Figure 1 show the micrographs of a one-time-deposited EDT-bridged PbS QDHN with different magnifications, revealing its hierarchical microporous and nanoporous structures. Sizeable voids of around 100−200 nm were uniquely formed in the monolayer. However, smaller voids, with an average size of ∼27 nm between clusters of QDs, would play a more critical role in increasing the QD accessible surface area. Furthermore, the interacting surface area of QDs will also be determined by the interdot distance in the nanopores. Therefore, it is essential to determine the nanopores' porosity to predict their supercapacitive EDLC performances.[29]

Accurately quantifying the porosity of nanoporous structures within thin films continues to be a significant and enduring challenge in fundamental research.[30,31] Using several available state-of-the-art approaches (Figure S3), we obtained a porosity of monolayer QDs of 73%, which is adequately high. A similarly porous monolayer formed on top of this monolayer after the LbL process would retain the same degree of porosity. The exact porosity value could be substantially higher given that the current consideration encompasses only the nanopores. Four neighboring QDs bridged by the EDT ligand will form a pore with a cross-sectional diameter of 2.1 nm, which is large enough for molecular ions such as $[EMIM]^+$ or $[TFSI]^-$ (size < 1 nm) to pass through.[32]

The total capacitance of the supercapacitor is dependent on the electrode thickness. To increase the capacitance of the microsupercapacitor, we continue the LbL dip-coating process to 5 layers (5L), 10 layers (10L), and 20 layers (20L). As additional dipping processes are conducted, subsequent QDs layers form a similar porous film while creating empty voids within the lower layer, developing a hierarchical nanoporous structure (Figures S4 and S5). AFM images reveal that the iodide-treated sample exhibits a denser porous structure featuring a macropore size of 0.7 μm. Moreover, the EDT ligands in the iodide-treated sample still govern the interdot

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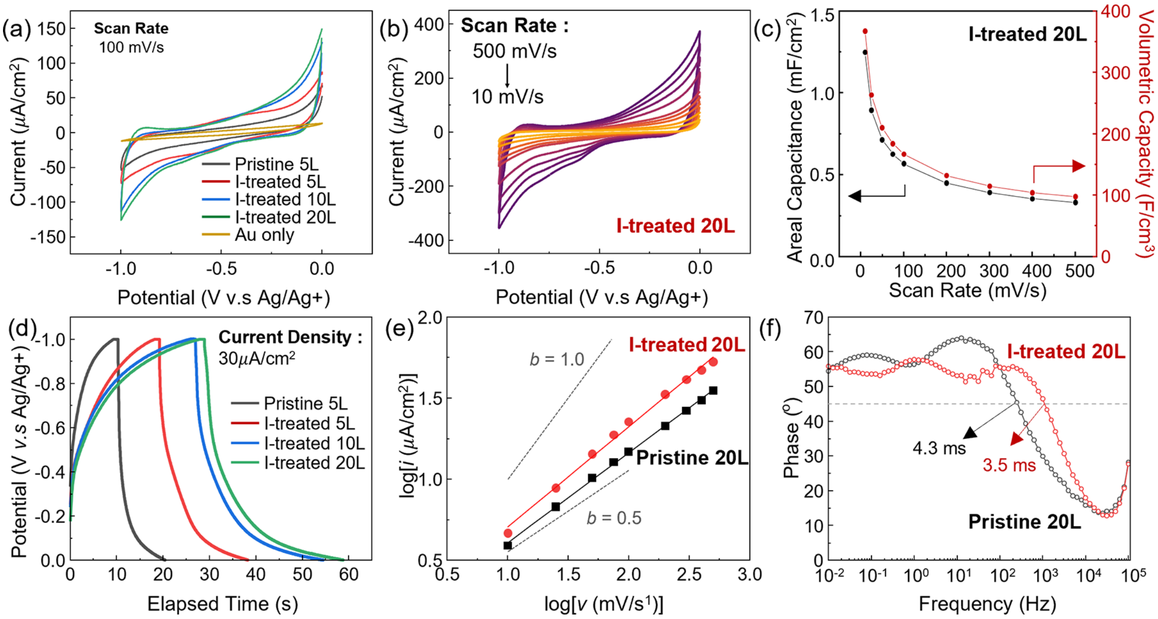

Figure 2. (a) Schematic CV curves of different QDHN treatments at a scan rate of 100 mV s^{-1}. The yellow line shows the CV curve of a bare current collector (gold, Au) for reference. (b) CV curves of iodide-treated QDHNs at various scan rates and (c) the corresponding extracted areal and volumetric values. (d) GCD curves of samples with different QDHN electrode thicknesses and treatments. (e) Plot showing the linear correspondence between the logarithm of anodic currents (i) and the logarithm of the scan rate at 0.5 V (vs Ag/Ag^+) of two different samples to elucidate the charge accumulation mechanism. (f) Bode plot obtained from EIS of two 20L samples with different treatments.

distances. At the same time, it results in thinner samples, almost half of the pristine (EDT-ligand-exchanged) sample. The use of iodide treatment also dramatically supports the controllability of the QD deposition.

3.2. Electrochemical Performance of the QDHN Electrode. Before using the QDHN as the electrode within the complete supercapacitor device, its charge storage mechanism was meticulously examined by using a three-electrode configuration immersed in pure IL. This process involved three distinct measurements: (i) CV, (ii) GCD, and (iii) EIS. Figure 2a shows the CV plots of various PbS QDHN electrodes in the potential window range between 0 and −1.0 V (vs Ag/Ag^+). The contribution of the current collector on the CV characteristics was negligible. We then compared the CV characteristics of the pristine EDT-bridged PbS QDHN (5L) and the iodide-treated EDT-bridged PbS QDHNs of various thicknesses. The quasi-rectangular shape of the CV plots of the pristine and iodide-treated QDHN electrodes shows no distinctive peak in the operating region for all scan rates (Figure 2b), indicating the EDLC characteristic of those electrodes without any indication of pseudocapacitive behavior.

Quantitatively, the EDLC behavior of these QDHN electrodes achieves unprecedented capacitance values. The 5L iodide-treated PbS QDHN electrode achieved an aerial capacitance value of 378 μF cm^{-2} at a fast scan rate (100 mV s^{-1}). This value is almost double the demonstrated value by the pristine PbS QDHN (197 μF cm^{-2}) (Figure S6). It should be noted that the ion size would limit the capacity of the QDHN to some extent.[33] Nevertheless, the PbS QDHN electrodes have almost equal capability to accumulate holes and electrons, suggesting their suitability to be applied as the electrodes of a symmetric supercapacitor.

Increasing the number of deposited QDHN monolayers increases the areal capacitance of the electrodes because the porous structure was retained. The areal capacitance of the 20L iodide-treated PbS QDHN electrode achieved a value as high as 1247 μF cm^{-2} (at 10 mV s^{-1}). When the obtained areal capacitance value was normalized by the thickness of the QDHN electrode, a volumetric capacitance of 366 F cm^{-3} was achieved (Figure 2c). The volumetric capacitance value of QDHN is among the highest for state-of-the-art supercapacitor materials (Table S1).[11,20,34−38] Notably, this value was accomplished in an intrinsic, template-less single-compound system, which sets it apart from many other supercapacitor materials,[20,39] which typically require other conducting matrices or backbones. The conduction of holes and electrons within these PbS QDHN hinges exclusively on the charge-transport process between the QD building blocks.[25,27,40−42]

Iodide-treated PbS QDHNs consistently outperform their untreated counterparts across all scan rates, displaying higher capacitance and superior rate capability (45% from 10 to 100 mV s^{-1}). The sustained rectangular shape of the curves, even at slow CV scan rates (10 mV s^{-1}) (Figure S6), signifies the well-preserved EDLC condition in both treated and untreated samples. The superior performance exhibited by the iodide-treated samples can be attributed to their larger, more defined pores. These refined pores enhance ion pathways, facilitating the increased diffusion of ions into the QDHNs, thus yielding improved performance metrics. Conversely, the enhanced conductivity in iodide-treated QDHNs also bolsters ion diffusion. The benefit of larger pores becomes more apparent at elevated scan rates, leading to capacitance values that remain relatively stable across both high and low scan rates. This observation endorses the hypothesis that the QDHN structure is pivotal in achieving a high charge-carrier density. Furthermore, iodide treatment effectively resolves the previously encountered limitation of accommodating rapid ion transfer.

The capacity of the QDHN was also characterized using GCD to confirm the robustness of the EDLC behavior (Figure 2d). GCD measurements were performed using a three-

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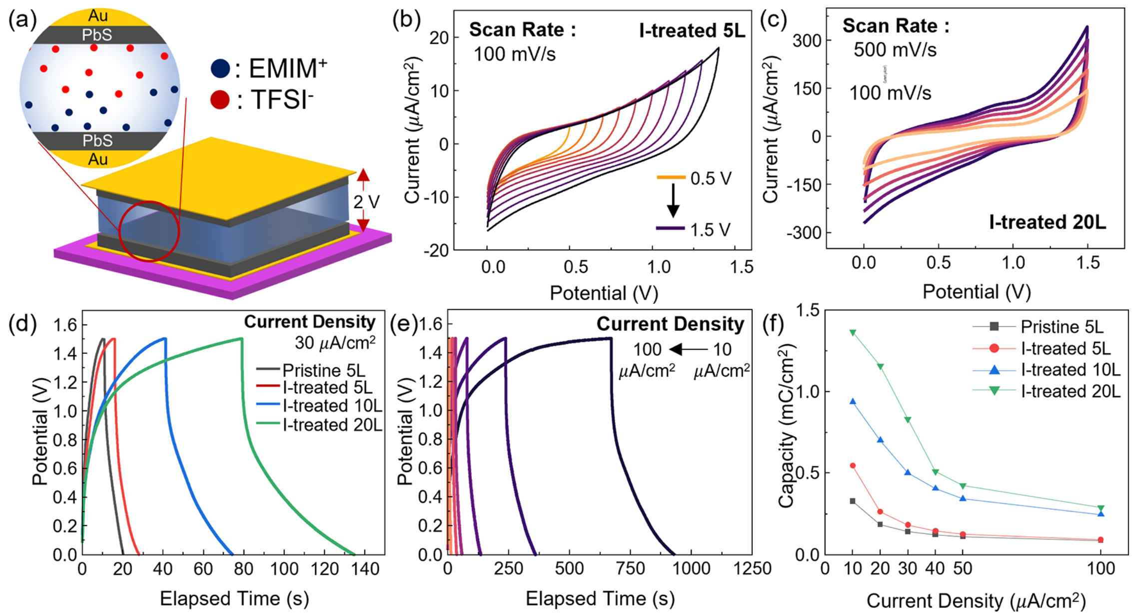

Figure 3. (a) Schematic illustration of a complete QDHN symmetric supercapacitor. (b) Measured CV plots showing the stable operational potential window of the complete supercapacitor device. (c) CV curves of the supercapacitor that utilizes 20L iodide-treated electrodes, which were measured at various scan rates. (d) GCD characteristics of the complete supercapacitor of QDHN electrodes with different thicknesses and treatment variations. (e) GCD characteristics of the supercapacitor with a 20L iodide-treated QDHN electrode under various charge and discharge current density values. (f) Extracted specific capacity of the complete supercapacitors with different variations of the QDHN electrodes.

electrode system with an applied current density ranging from 20 to 100 μA cm^{-2}. Figure 2d shows an almost linear GCD curve with no observation of an additional plateau, confirming the EDLC behavior of the QDHN.[43,44] The results are consistent for all four types of fabricated QDHN electrodes. Conversely, the nonlinear behavior at the end of the GCD cycles was expected due to the combination of modest ionic mobility of the neat IL and the QDs' internal resistance.[44,45] The 20L iodide-treated PbS QDHN electrode exhibits a specific capacity of 1.20 mC cm^{-2} (at 20 μA cm^{-2}) or equal to 296 C g^{-1} (at 5 A g^{-1}) (see Figure S3 for porosity determination). The 20L iodide-treated PbS QDHN shows the highest specific capacity among other samples (Figure S7). This value is exceptionally high for an EDLC. In agreement with CV measurements, the iodide-treated QDHN electrode consistently outperformed its untreated counterpart in terms of capacitance. Furthermore, QDHN electrodes with greater thickness were observed to have superior capacitance values.

To gain insights into the charge storage mechanism within these QDHN electrodes—whether it is governed by diffusion-controlled or surface-capacitive-controlled processes or a combination of both[46]—we conducted an electrochemical kinetic analysis of the CV and EIS measurement results. Figure 2e shows the relationship between the logarithm of the current density and the logarithm of the scan rate. Here, the obtained b value for the iodide-treated sample 20L is 0.61, which is higher than that of the untreated sample (b = 0.55). b values between 0.5 and 1.0 demonstrate a mixture of diffusion-controlled and surface-capacitive-controlled contributions on the QDHN capacitance properties.[19]

The surface-capacitive- and diffusion-controlled contributions were further evaluated (Figure S8). In the 5L iodide-treated sample, the observed capacitance appears predominantly to be influenced by the diffusion-controlled mechanism. However, in thicker samples, the surface-capacitive mechanism has an increased contribution. The 20L iodide-treated sample shows a rate-independent contribution of around 54% at 500 mV s^{-1}, higher than that of the untreated sample (40%). In regions with lower scan rates, ion diffusion serves as the primary controller of the QDHN capacitance, underscoring the crucial role of ionic mobility. This is evident because the iodide-treated QDHN film exhibits faster relaxation time (τ_0 = 0.8 ms) than the untreated sample (3.8 ms), indicative of enhanced ion movement and superior ion accessibility.[47] Extended ion paths are expected for thicker samples, potentially hindering ionic mobility. However, further iodide treatment of the 20L samples offered sufficient modification to the QD assembly, facilitating faster ionic relaxation, as indicated by the lower τ_0 (Figure S9). This discovery underscores the crucial role of the ligand in facilitating nanopore formation, which, in turn, bolsters the ion mobility and enhances the electronic conductivity of the QDHN. Both of these benefits contribute to the remarkable capacitance exhibited by the QDHN. Given this impressive electrochemical performance, the iodide-treated QDHN electrode holds considerable promise for high-performance energy storage when integrated into a symmetric supercapacitor device.

3.3. Stable Behavior under the Influence of Extra Charge. Two 0.5 cm^2 PbS QDHN electrodes were assembled to realize the complete supercapacitor device. In this supercapacitor device, a diluted [EMIM][TFSI] IL (1.0 M ACN) was utilized. Diluting the IL decreases its viscosity, elevating its ionic conductivity and enhancing the supercapacitor response and overall performance. In order to inhibit solvent evaporation, two-electrode measurements were conducted in a sealed bath. Given the suitability of the QDHN electrodes for both anodic and cathodic applications, the symmetric supercapacitor can operate within an extended potential window, doubling the voltage applied in the three-electrode measurements. However, this range is still below the maximum voltage range of the IL. The electrochemical window

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of the symmetric devices was then characterized by varying the voltage window in the CV measurement from 0.5 V up to 1.5 V (Figure 3b).

Even at 1.5 V, the QDHN electrodes maintain a consistent rectangular shape without any redox peak, affirming their excellent EDLC behavior. Furthermore, the symmetric supercapacitor demonstrates a robust capacitive performance across various scan rates, as evidenced by the sustained quasi-rectangular shape observed in CV. Notably, when neat IL was used, the QDHN electrodes could operate well even at higher voltage. A maximum voltage of 2.0 V is achieved, with the CV and GCD measurements confirming the continued good capacitive behavior of the QDHN electrodes (Figure S10). This comprehensive evaluation result underscores the remarkable electrical capabilities and potential of the QDHN-based supercapacitor system.

In agreement with the three-electrode measurement, the thicker sample has higher areal capacitance, as indicated by GCD measurements (Figure 3d,e). The 20L supercapacitor demonstrated a 1.36 mC cm^{-2} capacity at 10 μA cm^{-2}. This value is equivalent to 379 C cm^{-3} (at 2.7 A cm^{-3}) on the volumetric scale. This value is 1.2 times higher than the pristine sample under the same conditions, which achieved only 1.12 mC cm^{-2} (200 C cm^{-3}). Meanwhile, in the neat IL electrolyte, the sandwiched 20L of the iodide-treated QDHN has a capacity of up to 6.2 mC cm^{-2}, higher than that of the EDT-capped samples. This iodide-treated sample can hold up to 21% of its capacity at 100 μA cm^{-2}, while the pristine sample only holds around 17%. The enhanced capacitance observed in the iodide-treated devices can be attributed to the improved ion diffusivity within the electrodes.

Using a neat IL allows for a broader range of accessible potential windows, enabling higher power density. However, utilizing neat IL-based electrolytes in most supercapacitors often presents challenges associated with low ionic mobility, leading to compromised capacity retention.[48] Therefore, a less concentrated IL with lower viscosity is preferred for better capacity retention. Figure 4a shows the charging−discharging performance at the current density of 100 μA cm^{-2} over 5000 cycles. Capacity retention at around 80% was achieved in the supercapacitor that used 10L QDHN electrodes. The remarkable stability exhibited by QDHN highlights its ability to form robust bonds with the gold substrate, even without any binding agent. The strong adhesion can be attributed to the bidentate thiol group in the EDT ligand, which attaches to both the QD surface and the gold substrate.[49] Additionally, this finding suggests that, in the iodide-treated QDHN, the quantity of EDT on the QD surface remains substantial, serving as advantageous anchoring moieties.

The decrease in the capacitance value during charging and discharging primarily arises from issues related to the movement of the IL ions rather than the commonly observed degradation of electrode materials in other supercapacitor systems.[44,45] To validate this assertion in the context of QDHN supercapacitors, charging−discharging measurements were conducted with an extended resting time of 30 s after each cycle for 1000 cycles. We observed an increased capacitance value following the resting period. This indicates that the QDHN supercapacitor regains performance as the IL returns to equilibrium. This occurrence was never observed in standard (continuous) charging−discharging cycling. Furthermore, voltage scans revealed a voltage jump after the charging−discharging cycle, signifying the continued movement of ions even in the absence of applied current (Figure S11). These findings imply that an IL with a faster response would be necessary to accommodate thicker QDHN electrodes better and potentially achieve an even higher performance.

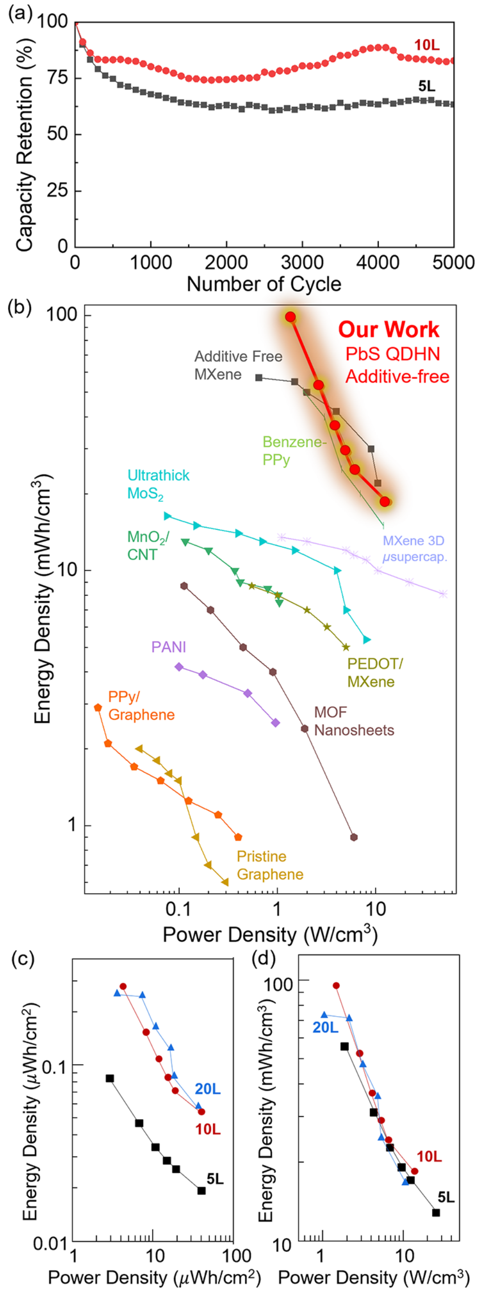

Figure 4. (a) Cycling performance of the QDHN-based supercapacitor devices with different numbers of (iodide-treated) layers of electrodes showing stable capacity retention. (b) Ragone plot comparing the volumetric performance of the QDHN-based supercapacitor (10 ML, iodide-treated) with the state-of-the-art supercapacitors. (c) Areal Ragone plot (energy density vs power density) of the supercapacitors that use different numbers of iodide-treated QDHN layers as the electrodes and (d) the corresponding volumetric Ragone plot signifying the thickness-independent volumetric performance.

3.4. Performance Benchmark. To demonstrate the superiority of our QDHN supercapacitor devices, we compared their performance with that of other electric energy storage devices. The comparison is represented in a Ragone plot, which correlates the energy density and power density of these devices. Both areal and volumetric densities can represent the energy density and power density metrics of energy storage devices (eqs 3 and 4).

Figure 4b shows the Ragone plot comparing the performance of the developed QDHN supercapacitor and those of the state-of-the-art supercapacitor materials, represented by the volumetric energy and power density. The volumetric energy density of the 10L iodide-treated QDHN supercapacitor achieved 95 mWh cm^{-3} (equivalent to 60 Wh kg^{-1}; Figure S12). The maximum power density of the QDHN reached 13.57 W cm^{-3} while maintaining an energy density of 18.51 mWh cm^{-3}. Furthermore, these QDHN supercapacitor devices outcompete the other high-performance supercapacitor materials, including MXene,[39,50−52] MoS_2,[53,54] Cl-doped graphene,[55] pristine graphene microsupercapacitors,[56] metal−organic frameworks,[57] polyanilines,[58] polypyrroles,[59,60] and MnO_2.[23]

From the areal energy density viewpoint only, a thicker electrode should demonstrate a higher energy density (Figure 4c). Ideally, the capacitance of the supercapacitor electrode increased linearly with the thickness of the sample. However, at the same time, the resistance and ion pathway length would also rise as the film gets thicker. This is the origin of why a thickness-independent capacitance for supercapacitor devices is so far hard to achieve, including in the pristine EDT-capped QDHN devices (Figure S13), and usually can only be observed in a system with very high ionic mobility or very conducting electrode materials.[16] These iodide-treated QDHN-based supercapacitors demonstrate coveted thickness-independent energy and power storage properties by showing similar volumetric characteristics for all different QDHN thicknesses (Figure 4d). This observation suggests that the iodide treatment has successfully addressed the low conductivity issue in thicker samples and allowed device scalability.

The ultrahigh volumetric energy density values achieved were enabled by the capability to arrange the small diameter QDs to overcome the low packing and low ionic mobility in the hierarchical nanoporous structure. In comparison, other nanoparticles that are relatively the same size as our QDs have a lower energy density (Table S2). Notably, the QDHN does not require any additive agent because we modulated the carrier conductivity in the assembly by tuning the distance of the QDs through ligand exchange and surface modification. So far, other QD-based or NP-based supercapacitors suffer low conductivity due to the long ligands that provide their solubility, thus demanding additional conducting supporting agents. Using these supporting agents sacrifices the merit of QD itself, which is the exploitation of its low dimensionality on both enhancement of the surface-to-volume ratio and the further utilization of quantum confinement effect phenomena.

On the other hand, the absence of a supporting conduction agent in these QDHNs still limits their Coulombic efficiency, affecting their cyclic performance. However, iodide treatment can significantly mitigate this (Figure S14). The modest electrical conductivity in the QDHN is thought to be the origin of the voltage drop during the discharging process of the supercapacitors (Figure 3d,e). The electronic coupling between the neighboring QDs that govern the conductivity is only controlled by the interdot distance, modulated using EDT ligands. Furthermore, the first layer of QDs is only attached to the gold current collector via the coordination bond of the thiol moieties of the functionalized EDT ligands on the QDs to the gold surface. Therefore, conduction of the QDHN is only governed by the above two factors, which quantitatively might be lower than systems that use highly conducting binding agents. This low Coulombic efficiency also originated from low ionic mobility due to the comparable size of the anions and cations of the ILs to the "last-nanometer" pore size. Replacing the electrolyte with faster ILs and enhancing the conductivity of the QDHN by controlling the electronic coupling between the QDs will further advance the development of high-performance energy storage devices.

4. CONCLUSIONS

In conclusion, we have successfully demonstrated the potential to develop high-performance additive-free supercapacitors with exceptional performance. This supercapacitor is based on hierarchical nanopores constructed from colloidal semiconducting QDs exhibiting unprecedented energy and power density levels. The utilization of highly uniform QD building blocks and meticulous control over ligand exchange and surface modification to enhance the conductivity enable the formation of hierarchical nanopores, leading to an extraordinary capacitance value. The well-engineered hierarchical nanoporous structure, tailored with specific pore sizes, offers significant advantages by facilitating rapid ionic transport while maintaining a high surface area for optimal accessibility. This innovative approach to developing additive-free materials extends beyond semiconducting colloidal QDs. It can also be applied to similarly sized metallic nanoparticles. Further advancements in QDHN supercapacitors can be achieved by implementing superior electrolytes and more robust QD compounds, harnessing quantum confinement effects, and scaling production processes. These endeavors will accelerate the realization of a carbon-neutral society through widespread electrification.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.4c02517.

Porosity estimation, FTIR absorption spectra of the QDHN, thickness measurement of the QDHN, details of the supercapacitor performance, specific areal capacity, contributions of surface-capacitive- and diffusion-controlled processes, Bode phase plots, gravimetric Ragone plot, areal Ragone plot, Coulombic efficiency comparison, and summary of the data plotted for Figure 4d (PDF)

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Author Contributions

S.Z.B. and F.I. conceived and conceptualized the project. M.A.I. performed most of the supercapacitor experiments. R.D.S. and S.Z.B. established the assembly methods. M.A.I. and R.D.S. performed the structural characterizations using AFM and TEM. R.D.W. and Y.M. obtained the assembly structure characterizations using FESEM. R.D.W. measured the charge-carrier transport of the assembly. M.A.I, R.D.S., F.I., Y.I., and S.Z.B. analyzed and discussed the experimental data. M.A.I., F.I., Y.I., and S.Z.B. wrote the manuscript with input from the other coauthors.

Notes

The authors declare no competing financial interest.

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